

**Rebuttal changes:**

**Comment 1: Reframe as journal article**

- **Action Taken:** The manuscript has been reframed as a journal article. All instances of the word "chapter" have been removed, specifically from the abstract and introduction.
- **Section Updated:** Abstract, Section 1

**Comment 2: Precise research problem**

- **Action Taken:** We explicitly added the Research Problem, Research Questions, and Research Objectives as structured bullet points to clarify the study's scope and intent.
- **Section Updated:** Sections 3.1, 3.2, 3.3

**Comment 3: Clarify novelty**

- **Action Taken:** We explicitly stated the technical novelty, emphasizing how the framework uses a continuous feedback loop via a Kafka event bus where threat intelligence dynamically updates the RL policies without human orchestration.
- **Section Updated:** Section 3.4

**Comment 4: Define Threat Model**

- **Action Taken:** The threat model has been expanded to specify external/internal threats and explicitly states the defender's assumed visibility over network flows and honeypot interaction logs via Kafka.
- **Section Updated:** Section 3.5

**Comment 5: Formal System Model**

- **Action Taken:** The system has been formally modeled as a decentralized, partially observable Markov Decision Process (MDP), defining the state space, action space, and continuous feedback loop.
- **Section Updated:** Section 4.1.1, Section 4.5

**Comment 6: Map HoneyBee concepts**

- **Action Taken:** The mapping between HoneyBee biological behavior (Scout Bees, Guard Bees, Waggle Dance) and cyber defense functions (Monitoring, Detection, Intelligence sharing) has been formalized into a clear grid table.
- **Section Updated:** Section 4.1 (Table 2)

**Comment 7: Revise Figure 2 (Diagram Alignment)**

- **Action Taken:** Figure 2 (Detect-Mislead-Neutralize-Learn Workflow) was updated to precisely match the text terminology, explicitly identifying the Detection, Deception, Response, and Learning layers.
- **Section Updated:** Section 4.8

**Comment 8: Revise Figure 2 (Diagram Components)**

- **Action Taken:** The components in the architecture diagram were revised to correctly display the Adaptive Honeypot, Reinforcement Learning engine, and Threat Intelligence Storage.
- **Section Updated:** Section 4.8

**Comment 9: Define Agent Roles**

- **Action Taken:** We explicitly defined the exact inputs, processes, and outputs of all four specialized agents (Monitoring, Detection, Deception, and Response) and how they coordinate over the Kafka publish-subscribe protocol.
- **Section Updated:** Section 4.7

**Comment 10: Specify ML Components**

- **Action Taken:** The specific hyperparameters and configurations for the machine learning components (XGBoost, CNN, LSTM, Adam optimizer) have been explicitly detailed in a dedicated table.
- **Section Updated:** Section 4.3 (Table 3)

**Comment 11: RL Formulation**

- **Action Taken:** The Reinforcement Learning (DQN) mechanism has been formally defined, detailing the State Space, Action Space, Reward Function, and the Bellman Equation policy update rule.
- **Section Updated:** Section 4.5 (Table 4)

**Comment 12: Adaptive Honeypot**

- **Action Taken:** The honeypot mechanism has been thoroughly explained as a Custom Application-Level Flask Deception Engine on port 5001, replacing previous generic references.
- **Section Updated:** Section 4.4

**Comment 13: 'Corneya' Typo**

- **Action Taken:** Typographical errors regarding honeypot terminology have been corrected and aligned with the custom application deception engine implementation.
- **Section Updated:** Section 4.4

**Comment 14: Dataset details**

- **Action Taken:** Full details of the dataset have been added in a table, confirming the use of the CIC-IDS2017 dataset, the 78 network-flow attributes, and the 70/15/15 train/validation/test split.
- **Section Updated:** Section 5.1 (Table 6)

**Comment 15: Experimental testbed**

- **Action Taken:** The hardware and software environment details for the experimental testbed were added, specifying Windows OS, Apache Kafka, ZooKeeper, and Python virtual environments.
- **Section Updated:** Section 5.2 (Table 7)

**Comment 16: Rebuild Table 2 (Main Results)**

- **Action Taken:** The main performance results table has been completely rebuilt to reflect actual empirical metrics rather than conceptual data.
- **Section Updated:** Section 5.3 (Table 8)

**Comment 17: Define evaluation metrics**

- **Action Taken:** The mathematical formulas for calculating Accuracy, Precision, Recall, and F1-Score based on the True Positive / False Positive confusion matrix have been explicitly added.
- **Section Updated:** Section 5.3

**Comment 18: Report precise metrics**

- **Action Taken:** Exact metrics have been reported, including an Accuracy of 98.24%, FPR of 1.85%, Latency of 420 ms, and Deception Engagement of 92.40%.
- **Section Updated:** Section 5.3 (Table 8)

**Comment 19: Statistical Reliability**

- **Action Taken:** To prove statistical reliability, the performance results are now reported as the mean performance over 10 independent runs, complete with 95% Confidence Intervals.
- **Section Updated:** Section 5.3 (Table 8)

**Comment 20: Baseline Comparison**

- **Action Taken:** A comparative benchmarking table has been introduced to compare the proposed system directly against existing baseline methods (DIAMoND, RL-IDS, Deep IDS).
- **Section Updated:** Section 5.7 (Table 9)

**Comment 21: Fair Comparisons**

- **Action Taken:** The baseline comparison was standardized to ensure all methods were evaluated on the exact same dataset metrics (Accuracy, FPR, Mean Time to Respond) for fairness.
- **Section Updated:** Section 5.7 (Table 9)

**Comment 22: Ablation Experiments**

- **Action Taken:** A new Ablation Studies section and table were added to empirically prove that disabling the RL response or deception layer degrades system accuracy and increases false positives.
- **Section Updated:** Section 5.8 (Table 10)

**Comment 23: Evaluate robustness**

- **Action Taken:** A note was added evaluating system robustness, confirming the framework maintained over 95% accuracy even when novel synthetic attack signatures (concept drift) were introduced.
- **Section Updated:** Section 5.2

**Comment 24: Implementation vs Conceptual**

- **Action Taken:** We explicitly clarified that the framework is fully implemented and operational via real-time Kafka streaming in a virtualized testbed, validating KPIs experimentally.
- **Section Updated:** Section 5.2

**Comment 25: Align conclusion**

- **Action Taken:** The conclusion has been softened to align directly with the empirical results, stating the framework "demonstrates significant potential" rather than making absolute claims.
- **Section Updated:** Section 7

**Comment 26: Strengthen Literature Review**

- **Action Taken:** The literature review has been strengthened by deeply integrating and discussing 21 peer-reviewed sources categorized across Machine Learning, Deep Learning, RL, and Deception.

- **Section Updated: Section 2**

**Comment 27: Verify References**

- **Action Taken:** The reference section was completely overhauled. Fabricated references were removed, and 21 verified academic sources formatted in Elsevier numbered style were added.
- **Section Updated: Section 2, References**

**Comment 28: Remove 'Proposed Framework'**

- **Action Taken:** The "Proposed framework" row was removed from the Literature Review summary table to ensure the table only summarizes prior literature.
- **Section Updated: Section 2 (Table 1)**

**Comment 29: Standardize terminology**

- **Action Taken:** The manuscript was thoroughly reviewed to ensure consistent capitalization and terminology, exclusively using the phrase "Detect-Mislead-Neutralize-Learn".
- **Section Updated: Throughout the manuscript**

**Comment 30: Improve research gap**

- **Action Taken:** The research gap identification was sharpened to explicitly note the lack of frameworks that autonomously link threat detection directly to adaptive deception for continuous learning.
- **Section Updated: Section 3.2**

**Comment 31: Algorithmic description**

- **Action Taken:** Algorithm 1 was added as a visual figure block, providing a formal pseudocode breakdown of the "Detect-Mislead-Neutralize-Learn" workflow executed for each network session.
- **Section Updated: Section 4.8**

**Comment 32: Discuss deployment risks**

- **Action Taken:** A new discussion paragraph was added acknowledging deployment risks, such as the potential for autonomous agents to block legitimate traffic, and recommending 'human-in-the-loop' approvals.
- **Section Updated: Section 6.2**

**Comment 33: Revise results discussion**

- **Action Taken:** The results discussion sections have been updated to explicitly reference specific tables (e.g., Table 7, Table 8) when quoting performance metrics.
- **Section Updated: Sections 5.1 to 5.8**